Every quasi-inverse of a function can be corrected into a half adjoint equivalence

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's labelled theorem thm:equiv-iso-adj at source lines 215-217; its proof environment starts on line 218, the line immediately after the statement environment ends, so the pairing is by adjacency and the retained proof range 218-242 is the whole environment.

Editorial note. The context span at source lines 212-213 is the author's own introduction to the theorem and carries no counter value.

Editorial note. The retained counters make the counted theorem print as Theorem 4.2.3, matching the original book numbering; the displays inside the proof are unnumbered, so nothing else consumes a counter value.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-equivalences/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: quasi-inverses can be corrected into half adjoint equivalences (author label thm:equiv-iso-adj, original Theorem 4.2.3).

Allowed standard prerequisites (editorial).

- Quasi-inverse. A quasi-inverse for f is a triple of a map g and homotopies from the composites to the identity maps.. It is the assumed datum (g, eta, epsilon) at source line 219.
- Half adjoint equivalence. A half adjoint equivalence is a quadruple of a map, its inverse and two homotopies together with a coherence path between the two ways of composing them.. It is the quadruple (g-prime, eta-prime, epsilon-prime, tau) that the proof has to produce at source line 220.
- Naturality of the two homotopies. The homotopies of a quasi-inverse satisfy naturality squares relating them to the maps on paths.. The two labelled naturality results invoked at source lines 232-235 are what make the coherence path computable.
- Cancellation in identity types. From a common right factor in an identity type one may cancel to obtain an equality of the two left factors.. It is the last step that turns the computed identity into the required coherence path at source line 241.

Source reference. Original corollary 2.4.4: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 180–180. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original lemma 2.4.3: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1540–1540. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Of course, it is obvious that $\mathsf{ishae}(f) \to \mathsf{qinv}(f)$: simply forget the coherence datum. The other direction is a version of a standard argument from homotopy theory and category theory.

Theorem 4.2.3. *For any $f : A \to B$ we have $\mathsf{qinv}(f) \to \mathsf{ishae}(f)$.*

Proof. Suppose that (g, η, ϵ) is a quasi-inverse for f. We have to provide a quadruple $(g', \eta', \epsilon', \tau)$ witnessing that f is a half adjoint equivalence. To define g' and η', we can just make the obvious choice by setting $g' :\equiv g$ and $\eta' :\equiv \eta$. However, in the definition of ϵ' we need start worrying about the construction of τ, so we cannot just follow our nose and take ϵ' to be ϵ. Instead, we take

$$\epsilon'(b) :\equiv \epsilon(f(g(b)))^{-1} \cdot (f(\eta(g(b))) \cdot \epsilon(b)).$$

Now we need to find

$$\tau(a) : f(\eta(a)) = \epsilon(f(g(f(a))))^{-1} \cdot (f(\eta(g(f(a)))) \cdot \epsilon(f(a))).$$

Note first that by original corollary 2.4.4, we have $\eta(g(f(a))) = g(f(\eta(a)))$. Therefore, we can apply original lemma 2.4.3 to compute

$$\begin{aligned} f(\eta(g(f(a)))) \cdot \epsilon(f(a)) &= f(g(f(\eta(a)))) \cdot \epsilon(f(a)) \\ &= \epsilon(f(g(f(a)))) \cdot f(\eta(a)) \end{aligned}$$

from which we get the desired path $\tau(a)$. □